SMS group

Safety provisions for danger zones
Fence elements

SN 401-2

ICS 13.110

Supersedes SN 401-2:2010-01

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1 Scope

This standard specifies the fence elements used in the SMS Siemag BA of SMS group consisting of fence nettings, posts, fastening elements and cover caps for use in fence systems as specified in SN 401-1.
The fence elements stated herein shall be used for the safety categories T I and T II (see SN 401-1).

2 Principles

For uniform indication of the dimensions in procurement drawings, the nominal dimensions 1400 and 2200 shall be used as height indications for the individual elements.
Unless otherwise indicated on the manufacturing drawings, the Manufacturing Instructions SN 200 apply to the manufacture of fence elements according to this standard.

3 Fence elements

Fence element is the general term for all fence nettings and fence posts.

3.1 Fence nettings

The protective fence system consists of frameless double-rod wire nettings with wire diameters of 8/6/8 mm (2 bars of 8 mm each in horizontal direction, 1 bar of 6 mm in vertical direction). The mesh width is 200/25 mm (200 mm horizontal spacing and 25 mm vertical spacing, see Fig. 1). To ensure resistance of the fence system against forces acting on it and against environmental influences, the welded fence nettings shall fulfill the requirements specified in DIN EN 10223-7, sections 6 and 7. Table 1 shows the sizes which are relevant for SMS Siemag. Whenever possible, fence nettings of 1000 and 2000 mm in width shall be used. Intermediate sizes shall be used only as adapting elements.

Edited by:
SMS Siemag
Standards Office

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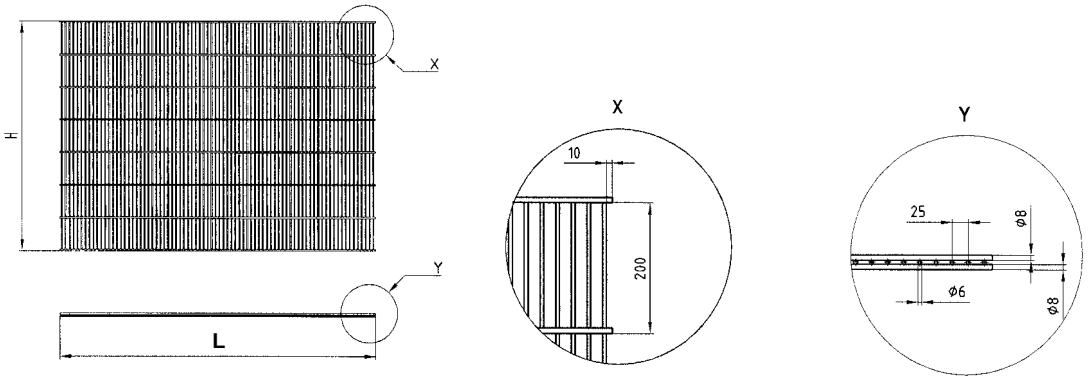


Fig. 1 – Fence netting DM

Table 1 - Dimensions

Ordering designation [a)]	**Height *H***	**Width *L***	**Weight in kg/unit**
SN 401-2 - DM1400x1000	1408	1000	7,98
SN 401-2 - DM-1400x2000	1408	2000	15,96
SN 401-2 - DM-1400	1408	Adapted [b)]	-
SN 401-2 - DM-2200x1000	2208	1000	12,54
SN 401-2 - DM-2200x2000	2208	2000	25,08
SN 401-2 - DM-2200	2208	Adapted [b)]	-

[a)] Nominal dimensions 1400 and 2200

[a] Given the space requirement of the retaining plates and for reasons of stability, the widths of the fence nettings to be adapted shall be in the range from 300 to 1975 mm and always with spacings of 25 mm.

3.2 Fence posts

The fence posts consist of square pipes with welded-on base plates. With the exception of types PR, PWR, PWL and PW, all fence posts are stiffened with gusset plates on the side of the danger zone. The top sides of the posts shall be closed with plastic cover caps, see item 4.
To achieve on fence systems according to SN 401-1 a connection of maximum stability between fence nettings and fence posts, the fence posts are provided with welded-on holding lugs. Fence posts can also be ordered as complete units provided with all necessary fastening elements.

3.2.1 Fence post elements

To allow flexible construction of the fence systems, fence posts are available in various basic types (see figures 2 to 6).
The detail views of the different types are shown in the figures 7 to 12. The dimensions are stated in Table 2.



Connecting post between fence elements

Fig. 2 – Type PV



End post as final element of a fencing, e.g. at a door.

Fig. 3 - Type PE



Right-angle corner post for the connection of two fence elements.

Fig. 4 –Type PR



Angle post, left, for the connection of fence elements at an angle of 45°.

Fig. 5.1 – Type PWL



Angle post, right, for the connection of fence elements at an angle of 45°.

Fig. 5.2 – PWR



T-post for the connection of 3 fence elements at an angle of 90°.

Fig. 6 - PT

Table 2 – Fence post elements

Type	Ordering designation[a)]	Height H	Distance C	Distance D	Distance E	Weight kg/unit
	SN 401-2-PV-1400	1410	370	508	380	9,94
	SN 401-2-PER-1400	1410	370	508	380	9,56
	SN 401-2-PEL-1400	1410	370	508	380	9,56
	SN 401-2-PR-1400	1410	370	508	380	10,58
	SN 401-2-PWL-1400	1410	370	508	380	10,58
	SN 401-2-PWR-1400	1410	370	508	380	10,58
	SN 401-2-PT-1400	1410	370	508	380	10,96
	SN 401-2-PV-2200	2210	570	908	580	14,45
	SN 401-2-PER-2200	2210	570	908	580	14,07
	SN 401-2-PEL-2200	2210	570	908	580	14,07
	SN 401-2-PR-2200	2210	570	908	580	15,09
	SN 401-2-PWL-2200	2210	570	908	580	15,09
	SN 401-2-PWR-2200	2210	570	908	580	15,09
	SN 401-2-PT-2200	2210	570	908	580	15,47

a) Nominal dimensions 1400 and 2200

3.2.2 Fence connecting post PV

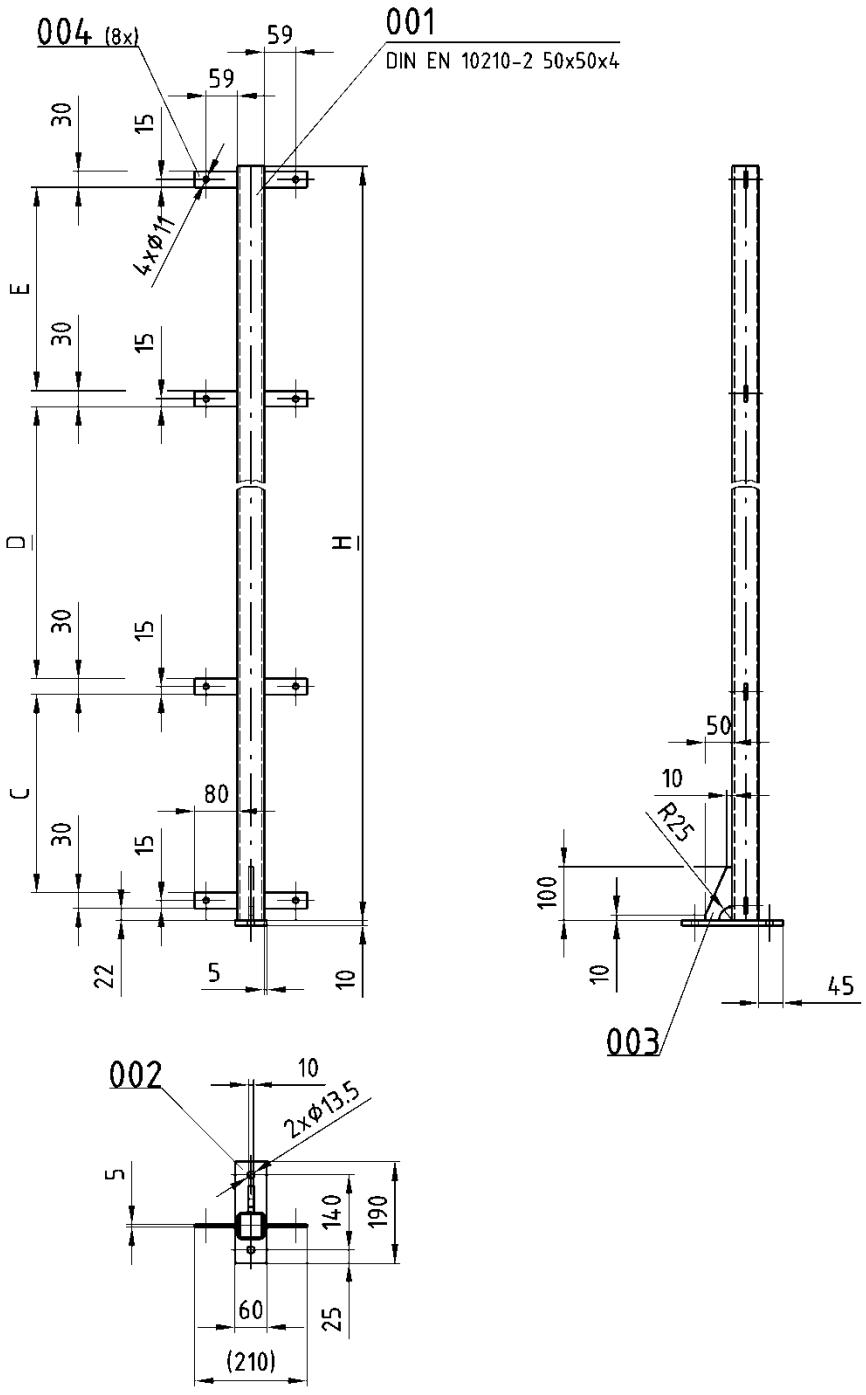


Fig. 7 – Fence connecting post PV

3.2.3 Fence end post PER

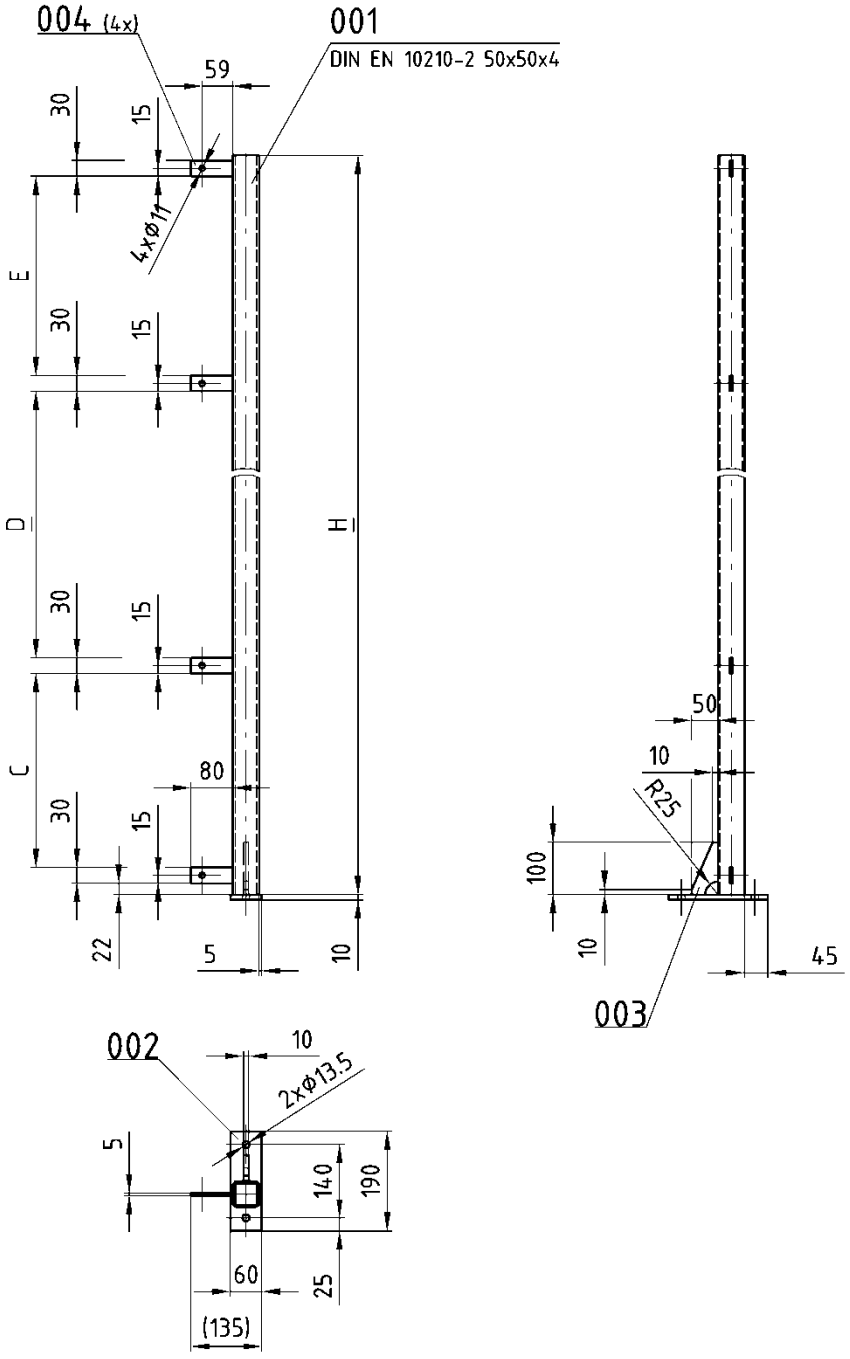


Fig. 8 – Fence end post PER

3.2.4 Fence end post PEL

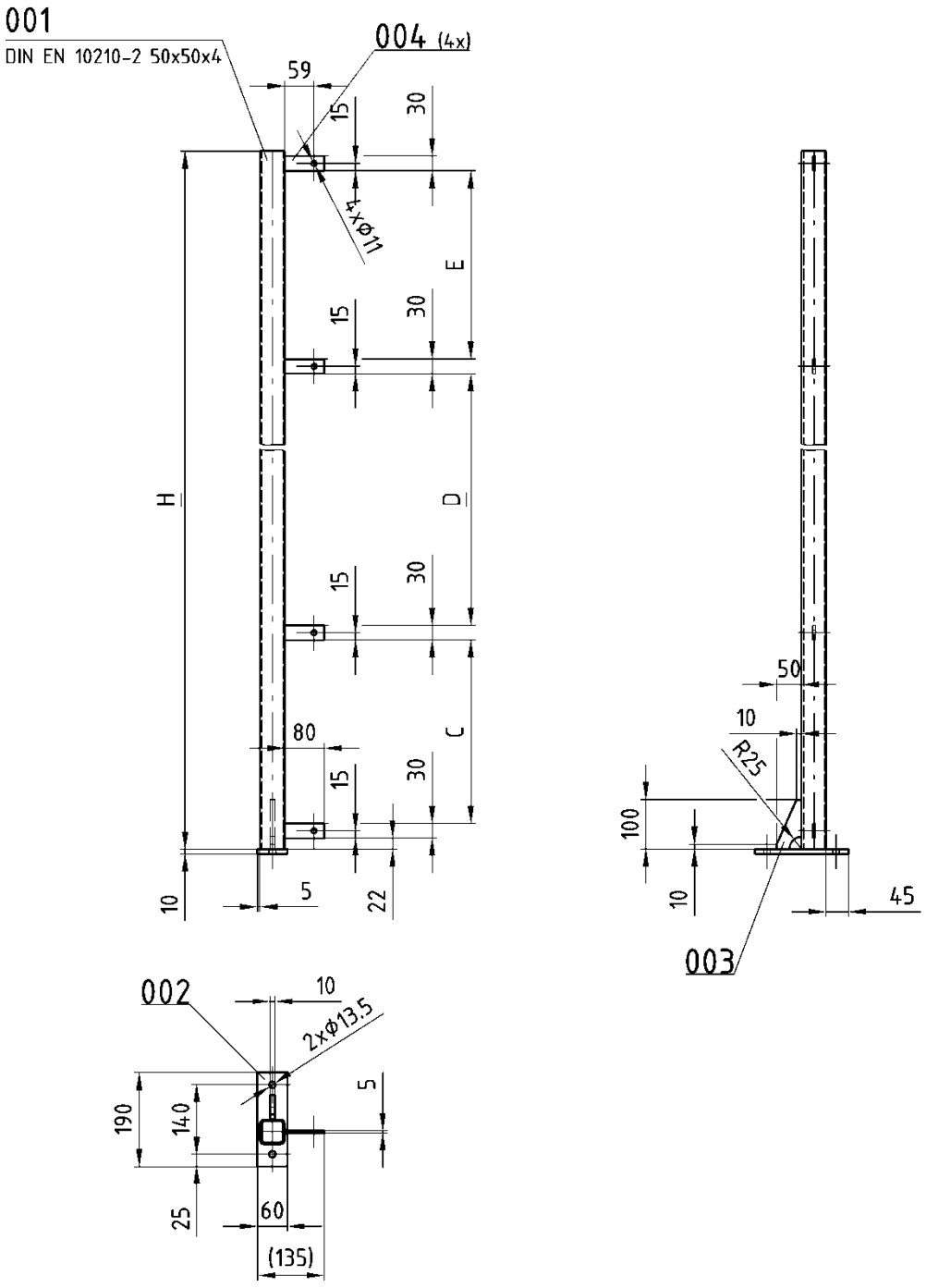


Fig. 9 – Fence end post PEL

3.2.5 Fence corner post PR

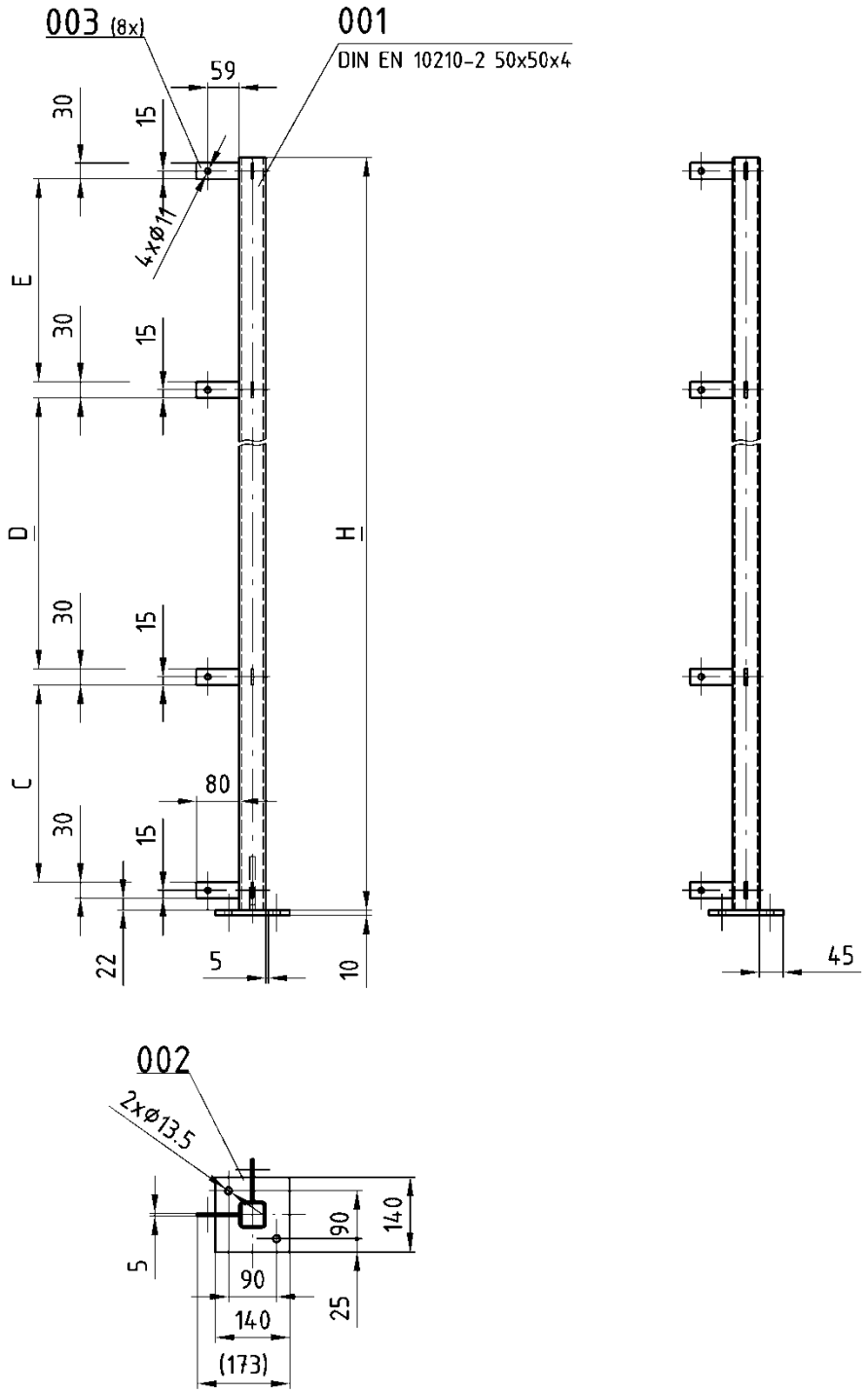


Fig. 10 – Fence corner post PR

3.2.6 Fence angle post, left, PWL

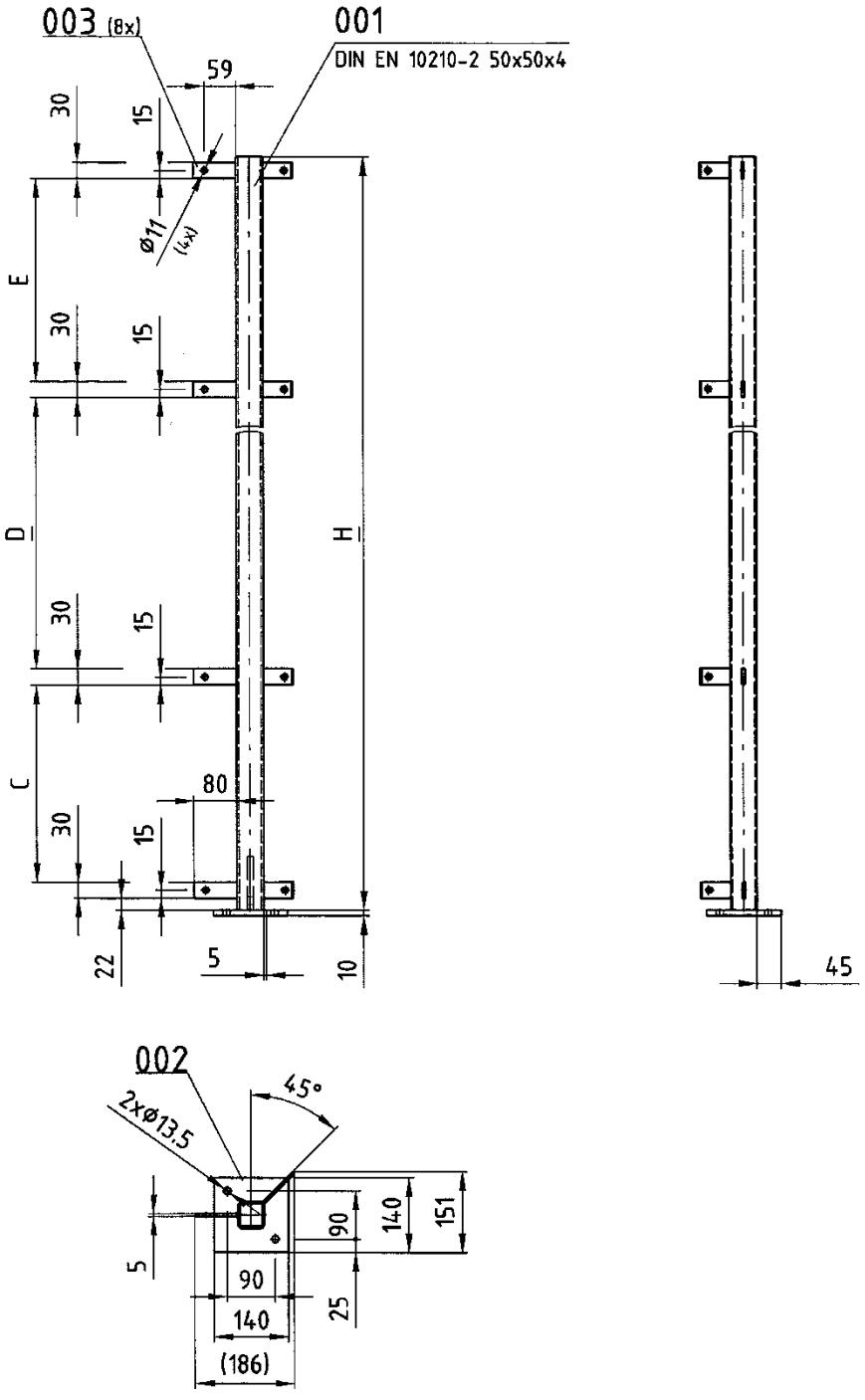


Fig. 11 – Fence angle post, left, PWL

3.2.7 Fence angle post, right, PWR

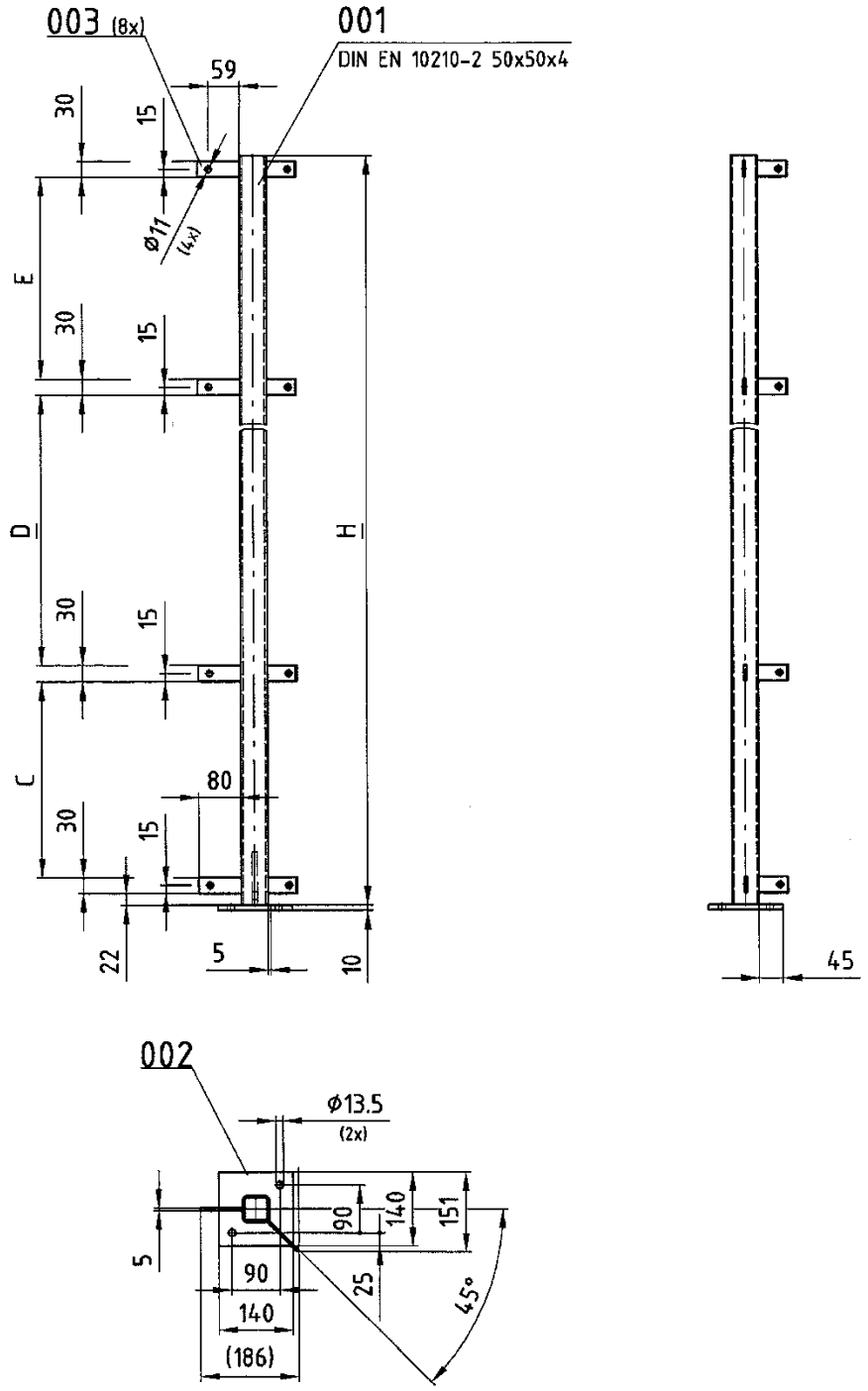


Fig. 12 – Fence angle post, right, PWR

3.2.8 Fence T-post PT

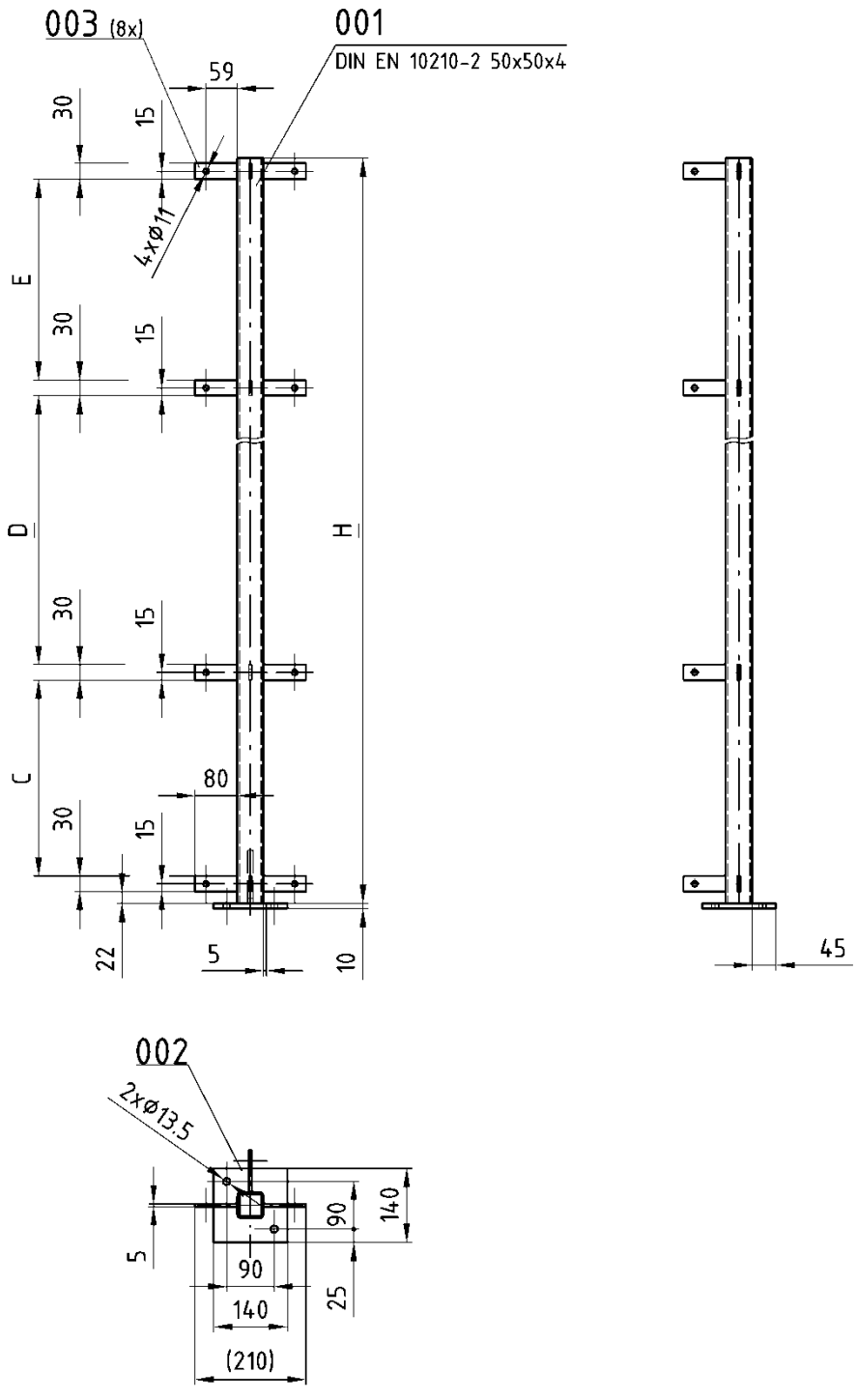


Fig. 13 - Fence T-post PT

4 Fastening elements

The fence and door posts are fastened in the floor with stud anchors available as catalogue parts according to Table 3.
The fence nettings are fastened to the posts with prefabricated retaining plates and mushroom-head square neck bolts (see figures 14 and 15) to prevent loosening of the connection from outside.
The fastening elements are shown in Table 3.
Welding up of fence elements during assembling or other ways of fastening are not allowed.

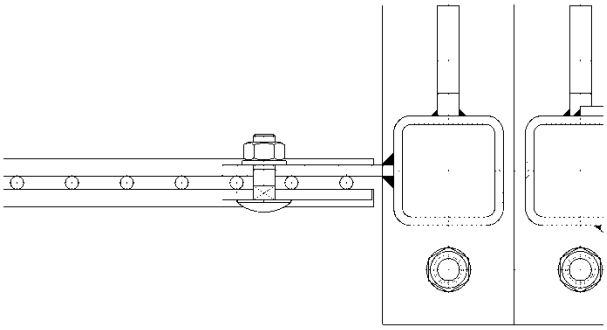


Fig. 14- Fastening

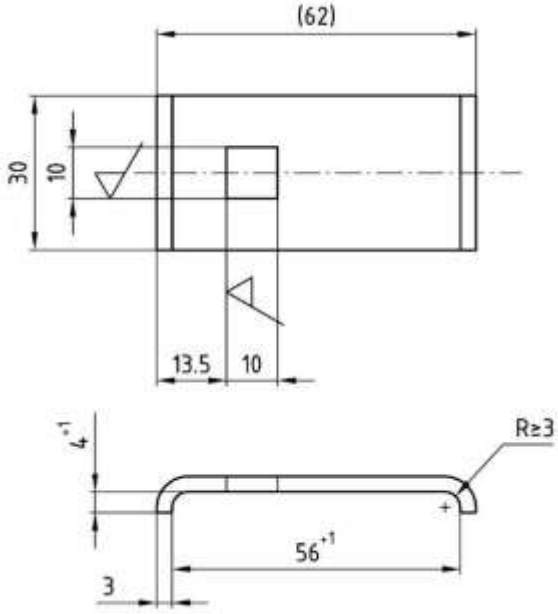


Fig. 15 - Retaining plate HB

Table 3 – Fastening elements for fence systems

Designation	Ordering indication	Material indication for fence system made of S235JR [a)]	Material indication for fence system made of X5CrNi18-10 (1.4301)	Weight kg/unit
Retaining plate	SN 401-2 - HB	S235JR	X5CrNi18-10 (1.4301)	0,075
Stud anchor	HILTI - HST- M12x115/20	St GLVZ	-	0,150
Stud anchor	HILTI - HST- HCR - M12x115/20	-	X1NiCrMoCuN25-20-7 (1.4529)	0,150
Mushroom-head square neck bolt	DIN 603 - M10x30	8.8 A0U	A2-70	0,025
Washer	ISO 7089 -10	ST-HV200	A2-200	0,004
Hexagon nut	ISO 4032 - M10	8 A0U	A2-70	0,012

[a)] Fence system made of S235JR (with subsequent powder coating) or S235JR TZN

5 Cover cap

To protect the posts in rough ambient conditions, every fence and door post according to SN 401-2 and 401-3 shall be provided with protective plastic cover (Fig. 16) at the top side. The design of the inner part of the cap is at the maker's choice and shall be suitable for pipe wall thicknesses of 4 to 6,3 mm. Data of the cover cap are shown in Table 4.

Table 4 – Cover cap

Designation	Ordering indication	Material [a)]	Colour	Weight kg/unit
Cover cap	SN 401-2 - AK	PE	Black	0,025
[a] Cover caps are always made of PE irrespective of the fence system material.				

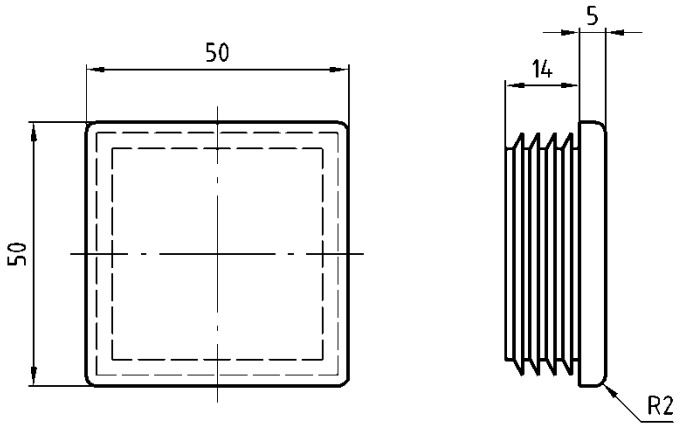


Fig. 16 – Cover cap AK

6 Fence post units

All fence posts stated in Table 2 can also be ordered as complete units in the material required and with all pertaining fastening elements and cover cap. To indicate the complete unit, the letter "E" shall precede the ordering indication of the respective fence post. Table 5 shows the bill of materials for a complete fence post unit as an example.

Table 5 – Bill of materials (example)

Quantity	Designation
1	Fence post unit SN 401-2-EPR-2200-X5CrNi18-10
Consisting of:	
1	Fence corner post SN 401-2 – PR – 2200 - X5CrNi18-10
1	Cover cap SN 401-2 - AK - PE
8	Retaining plate SN 401-2 - HB - X5CrNi18-10
8	Mushroom-head square neck bolt DIN 603-M10x30 - A2-70
8	Washer ISO 7089 – 10 St-HV200 - A2-200
8	Hexagon nut DIN EN ISO 4032 - M10 - A2-70
2	Stud anchor HILTI - HST - HCR - M12x115/20 - 1.4529

7 Materials

8 A0U	according to DIN EN ISO 898-6, galvanised
8.8 A0U	according to DIN EN ISO 898-1, galvanised
A2-70	according to DIN EN ISO 3506-1
A2-200	according to DIN EN ISO 3506-1
PE	according to DIN EN ISO 1043-1, polyethylene
S235JR	according to DIN EN 10025-2, structural steel, surface untreated (for subsequent powder coating)
S235JR TZN	according to DIN EN 10025-2, structural steel, surface galvanised,
St GLVZ	steel, electrogalvanised
X1NiCrMoCuN25-20-7 (1.4529)	according to DIN EN 10028-7, austenitic stainless steel, without surface treatment.
X5CrNi18-10 (1.4301)	according to DIN EN 10088-3, austenitic stainless steel, without surface treatment.

8 Examples of designation

Designation of a fence netting with 1400 mm nominal height, 1000 mm long, made of galvanised structural steel:

Fence netting SN 401-2-DM-1400x1000-S235JR-TZN

Designation of an adapted fence netting 1375 mm long, nominal height 2200 mm, made of X5CrNi18-10 (1.4301):

1375 mm fence netting SN 401-2-DM-2200-X5CrNi18-10

Designation of an angle fence post with 1400 mm nominal height, made of structural steel, untreated:

Fence post SN 401-2-PW-1400-S235JR

Designation of a fence end-post unit for left end, 2200 mm nominal height, and made of galvanised structural steel:

Fence end post unit SN 401-2-EPEL-2200-S235JR+N TZN

Designation of a cover cap:

Cover cap SN 401-2-AK

Referenced technical standards, codes and regulations

DIN 603	Cup head square neck bolts
DIN EN 10025-2:2005-04	Hot rolled products of structural steels - Part 2: General technical delivery conditions for non-alloy structural steels
DIN EN 10028-7:2008-02	Flat products made of steels for pressure purposes - Part 7: Stainless steels
DIN EN 10088-3:2005-09	Stainless steels - Part 3: Technical delivery conditions for semi-finished products, bars, rods, wire, sections and bright products of corrosion-resisting steels for general purposes
DIN EN 10210-2	Hot finished structural hollow sections of non-alloy and fine grain steels - Part 2: Tolerances, dimensions and sectional properties
DIN EN 10223-7	Steel wire and wire products for fencing and netting - Part 7: Steel wire welded panels for fencing
DIN EN ISO 898-1	Mechanical properties of fasteners made of carbon steel and alloy steel - Part 1: Bolts, screws and studs with specified property classes - Coarse thread and fine pitch thread
DIN EN ISO 898-6	Mechanical properties of fasteners - Part 6: Nuts with specified proof load values; Fine pitch thread
DIN EN ISO 1043-1	Plastics - Symbols and abbreviated terms - Part 1: Basic polymers and their special characteristics
DIN EN ISO 3506-1	Mechanical properties of corrosion-resistant stainless steel fasteners - Part 1: Bolts, screws and studs
DIN EN ISO 4032	Hexagon nuts, style 1 - Product grades A and B
DIN EN ISO 7089	Plain washers - Normal series, product grade A
SN 200	Manufacturing Instructions
SN 401-1:2011-08	Safety provisions for danger zones; Fence systems; Overview - Definitions
SN 401-3:2013-04	Safety provisions for danger zones; Door units and door elements

Revision of April 2013

Section on Principles added.
Retaining plate HB (Fig. 15) replaced.
Surface short designations revised according to SN 117.